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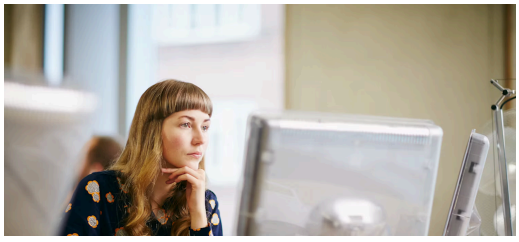
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How to Become a Cloud Architect: Degrees, Skills, Jobs, Courses

Written by Coursera Staff • Updated on Mar 24, 2026
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Learn how to become a cloud architect, including the skills it takes to succeed in this role, to figure out if this is the right direction for your career.



Key takeaways

To become a cloud architect, you'll need a strong background in computer operating systems, programming languages, networking, and security.

- The median total salary for a cloud architect is $200,000 [1].
- Common cloud architect roles include IT support technician, web developer, and CMS manager.

- You can become a cloud architect by earning a degree, gaining experience, and pursuing cloud architecture certifications.

Discover what's needed to build a successful career to help you decide if cloud architecture is the right path for you. Afterward, consider enrolling in the AWS Cloud Solutions Architect Professional Certificate. In as little as two months, you'll have the opportunity to make informed decisions about when and how to apply key AWS Services for compute, storage, database, networking, monitoring, and security. Upon completion, add this shareable credential to your resume or LinkedIn profile.



professional certificate

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Start here to become an AWS Solutions Architect. Gain the skills and knowledge to design architectural solutions on AWS and prepare for the AWS Certified Solutions Architect - Associate certification exam.

4.8

(7,134 ratings)

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Skills you'll build:

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Best Cloud Architecture courses for beginners

Top-rated beginner-friendly Cloud Architecture courses with no prerequisites.

1. 1
   Introduction to Cloud Computing
   IBMBeginner1 - 3 MonthsCourse4.6(7,771)
2. 2
   Cloud Architecture Design Patterns
   CourseraBeginner1 - 4 WeeksCourse4.7(36)
3. 3
   Architecting with Google Kubernetes Engine
   Google CloudBeginner3 - 6 MonthsSpecialization4.6(5,176)

What does a cloud architect do?

As a cloud architect, you'll have an important role in any IT department; you'll be tasked with designing, developing, implementing, and supporting cloud-computing solutions. Cloud computing is beneficial for businesses since it can potentially reduce business costs while enhancing innovation and improving compliance and security.

Cloud architects commonly report to IT directors, chief technical officers, or other members of upper management. In this role, you'll be expected to manage the organization's cloud database, develop new solutions to support the company's goals, and stay on top of all the latest trends and innovations in cloud computing. Common responsibilities include:

- Meeting with senior management and executives to discuss the company's long- and short-term goals and to help them understand how cloud-based solutions can help them meet those goals.
- Creating ongoing cloud strategies that align with organizational goals
- Planning the organization's cloud, designing the solutions and strategies, and developing cloud-based applications
- Evaluating and maintaining existing cloud applications, in addition to handling any technical problems that arise
- Monitoring and maintaining privacy in the cloud and ensuring that data, assets, and applications are secure
- Designing cloud security architecture and employing relevant tools
- Creating a culture of cloud adoption by creating a plan for unique cloud solutions, documenting cloud processes and procedures, training others on cloud solutions, and remaining open to ongoing change based on the latest data
- Designing and implementing the organization's cloud usage practices, including rules and standards

- Choosing cloud service providers based on criteria such as security practices, compliance with your industry, and how the architecture will integrate into the company's structure and workflows

Read more: How to Become a Cloud Security Engineer: Career Guide

Cloud architect salary and job outlook

As a cloud architect, many different factors may impact how much you'll earn, including the amount of experience, location, and the industry you work in.

Overall averages can give you a good idea of what you might expect. The median total salary for cloud architects is $200,000 [1]. The potential range is between $157,000 to $258,000 [1]. These figures include base pay plus additional pay, such as bonuses, profit-sharing, and commissions.

Salary at different levels: Is cloud architect a high-paying job?

Your experience level is one factor determining how much you can expect to earn as a cloud architect. Many cloud architects earn additional money beyond their base salary through commissions, tips, profit-sharing opportunities, and bonuses, which can boost your annual earnings at every level. The median total pay, which includes base pay plus additional pay, such as bonuses, profit-sharing, and commissions, for entry-, mid- , and senior-level cloud architects is as follows [1]:

- **0–1 years:** $137,000
- **4–6 years:** $161,000
- **10–14 years:** $186,000

What is the demand for cloud architect roles?

Cloud services are driving the digital transformation and are continuing to grow across a variety of sectors, including government, technology, health, education, and more.

The US Bureau of Labor Statistics (BLS) estimates the job growth rate for all computer network architects, including those who work in the cloud, to be 12 percent from 2024 to 2034 [2]. As more businesses and industries adopt cloud-based services, the number of job openings may continue to increase.

Required skills for cloud computing and architecture

Before getting started as a cloud architect, you'll need a strong foundation in cloud computing. This includes understanding different cloud service providers, best practices in cloud security, and types of cloud service models and deployment architectures. Developing a robust set of workplace and technical skills can help you prepare for this role. Helpful cloud architect skills include:

- **Enterprise computing:** You'll need to be able to work with different operating systems, including Linux, Windows, Unix, and Ubuntu.

- **Basic programming languages:** Familiarity with programming languages like Python, C#, or Java can help you better create solutions and proof of concepts to demonstrate how they'll work to senior executives.
- **Networking:** As a cloud architect, you should be able to work with concepts like Domain Name System (DNS), transmission control protocol/internet protocol (TCP/IP), virtual private network (VPN), and hypertext transfer protocol (HTTP). Without a good understanding of networking, creating effective cloud-based solutions that are both scalable and secure may be difficult.
- **Cloud-based security concepts:** Securing access and data is essential to your role as a cloud architect. This means you should be proficient in concepts like using Identity and Access Management, firewalls, VPN, and in-plane switching (IPS) systems.
- **Basic business skills:** As a cloud architect, you'll create business solutions to support the organization's goals, which means having a good foundation in business skills like calculating return on investment, how to read financial statements, and the basics of what C-level executives care about.
- **Leadership:** You'll often lead teams of other IT professionals, including developers and engineers, who are working with you to build strong, efficient cloud environments. Good teamwork, leadership, and communication are key.

What qualifications do you need to be a cloud architect?

Cloud architects often have a bachelor's degree in cloud computing or a similar field, like computer science, cybersecurity, or engineering. Many cloud architects earn certifications, generally to build their skills in vendor-specific solutions.



Common roles for beginning a career as a cloud architect

All salary data is the median total pay sourced from Glassdoor as of March 2026. These figures include base pay plus additional pay, such as bonuses, commissions, and profit-sharing.

There are a variety of potential career paths that may help you become a cloud architect. This role is usually a mid-to-senior-level position, requiring you to have work experience before getting started as a cloud architect. Many people begin their careers in roles such as:

- IT support technician**:** In this role, you'll support, install, troubleshoot, and maintain hardware and software for an organization.

- **Average annual salary (US):** $68,000 [3]

- **Cloud support associate:** In this role, you'll provide customer service in the cloud, helping you gain experience while also learning and using troubleshooting techniques, interacting with engineers, and supporting a positive customer experience by helping them solve issues.

  - **Average annual salary (US):** $108,000 [4]



professional certificate

Microsoft Cloud Support Associate

Launch your career as a Cloud Support Associate. Build job-ready skills for an in-demand career in information technology in as little as 6 months. No prior experience required to get started.

4.7

(1,685 ratings)

19,010 already enrolled

Beginner level

Learn More

Average time: 5 month(s)

Learn at your own pace

Skills you'll build:

Hybrid Cloud Computing, Microsoft Azure, Collaborative Software, Cloud Services, Cloud Solutions, Load Balancing, Microsoft Power Automate/Flow, Authorization (Computing), General Networking, Cloud Deployment, Authentications, Microsoft Teams, Hardware Troubleshooting, Security Controls, Generative AI, IT Automation, Dashboard Creation, Cloud Computing, Computer Networking, Network Protocols, Network Troubleshooting, Microsoft Windows, Technical Support, System Monitoring, Technical Documentation, Technical Communication, Operating Systems, Computer Hardware, Mac OS, Cloud Security, Disaster Recovery, Patch Management, Microsoft Copilot, Cloud Management, Virtual Machines, Microsoft Excel, Data Loss Prevention, Microsoft PowerPoint, Microsoft Outlook, Microsoft 365, Endpoint Security, Microsoft Office, Cloud-Based Integration, Productivity Software, Data Security, Cloud Storage, Virtual Networking, System Configuration, Network Security, Cloud Infrastructure, Role-Based Access Control (RBAC), Identity and Access Management, Application Deployment, Kubernetes, Network Architecture, Cloud Computing Architecture, Network Administration, Cloud Hosting, Network Monitoring, Data Storage, Data Management, Software Installation, Hardening, Information Systems Security, Servers, Business Software, Cybersecurity, Computer

Architecture, AI literacy, Computer Systems, Encryption, Multi-Factor Authentication, Virtual Private Networks (VPN), Network Performance Management, Information Privacy, Solution Design, Query Languages, Incident Response, Resource Management, Event Monitoring, Performance Tuning, Azure Active Directory, Infrastructure as Code (IaC), Windows PowerShell, Network Planning And Design, Command-Line Interface, OSI Models, Firewall, TCP/IP, Virtualization, Network Model, Networking Hardware, Network Infrastructure, Technical Support and Services

- **Web developer (back-end, front-end, or full-stack):** In this role, you'll handle all aspects of creating and maintaining websites, including creating and testing applications, writing code, and working with other IT teams to bring your company's or client's vision to life.
    - **Average annual salary (US):** $100,000 [5]
- **Cloud engineering intern:** In this role, you may be tasked with attending training and shadowing full-time cloud engineers. You'll also work on responding to customer tickets and learning to build cloud infrastructure.
    - **Average annual salary (US):** $145,000 [6]
- Cloud engineer**:** In this role, you'll assess the technological needs of your clients and create solutions. You'll be tasked with planning and creating, designing and testing, and managing and maintaining cloud computing applications.
    - **Average annual salary (US):** $151,000 [7]
- **CMS manager:** As a content management system manager, you will oversee metrics and manage a team to create content that helps a company develop, establish brand recognition, and meet company goals, like driving traffic to its website or increasing sales.
    - **Average annual salary (US):** $129,000 [8]
- **Web editor:** In this role, you'll create a website's editorial strategy and content, work with a creative team to develop the material, and keep the website up to date. You may also create polls or other interactive content.
    - **Average annual salary (US):** $92,000 [9]

Steps for how to become a cloud architect

Ready to get started? Although there aren't any formal education requirements to become a cloud architect, you will need to develop a robust set of relevant computing and cloud architect skills. These three steps can help you build your career.

1. Earn a bachelor's degree.

Most employers prefer hiring candidates with at least a bachelor's degree. Earning a degree in computer science or a related field is also one of the best ways to develop both theoretical and practical knowledge and

skills. It may give you a strong foundation in working with operating systems, understanding networking concepts, best practices in cloud security, and programming languages.

2. Gain professional experience.

A cloud architect role isn't an entry-level job. You need to gain on-the-job experience working in the technology sector. Because there's a lot of overlap in various IT roles, you can start your career in many different ways and remain on the path to your goal. Some common entry-level roles include cloud or IT support, IT support technicians, web developers, system engineers, database administrators, developers, and data management professionals.

3. Get one or more cloud architecture certifications.

Getting certified doesn't guarantee you'll get a job, but it can help make you more competitive in the job market and increase your earning potential. Certifications offer proof of your skills and expertise in that specific platform. Some popular cloud architecture certifications include:

- **Associate AWS Certified Solutions Architect:** You'll need at least one year of experience using AWS to design systems and the ability to design and deploy effective, secure applications using Amazon's cloud platform.
- **Associate AWS Certified Developer:** You'll need at least one year of experience using Amazon's cloud to develop and maintain applications. It's designed to demonstrate your proficiency in programming languages, AWS best practices, AWS core services, and writing code for serverless apps.
- **Microsoft Azure Solutions Architect Expert:** There are a few different levels of Azure certifications. Fundamentals cover the basics, while the administrator associate requires six months or more of experience with Azure administration, and the solutions architect expert requires extensive knowledge of using Azure for cloud development, platform design, and data security.
- **Google Professional Cloud (GPC) Architect:** This certification requires skills in using GPC to design and develop cloud-based solutions, along with managing, scaling, and securing them while adhering to compliance standards.
- **IBM Certified Solution Architect:** You'll need to demonstrate that you have a solid understanding of the architecture of IBM cloud infrastructure. You'll also need to be able to construct solutions to meet unique needs and apply the industry's best practices.
- **Arcitura Certified Cloud Architect:** This versatile credential demonstrates extensive knowledge of cloud platforms' structure and mastery of design principles, engineering practices, and hands-on application.

Pearson

specialization

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Prepare to Pass the CompTIA Cloud+ Exam. Get targeted training on objectives in the Cloud+ cert exam and learn core cloud computing skills.

5.0

(7 ratings)

349 already enrolled

Beginner level

Learn More

Average time: 1 month(s)

Learn at your own pace

Skills you'll build:

Network Troubleshooting, Security Controls, Solution Design, Cloud Computing, Data Security, Cloud Computing Architecture, Identity and Access Management, Network Security, Cloud Deployment, Cloud Security, Cloud Solutions, Vulnerability Assessments, Disaster Recovery, Application Lifecycle Management, Cloud Infrastructure, Cloud Management, Scalability, CI/CD, Incident Response, Infrastructure as Code (IaC), Cloud Storage, Data Migration, Virtual Private Networks (VPN), Virtualization, Data Storage Technologies, Network Routing, Data Storage, Patch Management, Encryption, Cloud Applications, Performance Tuning, Capacity Management, System Monitoring, Network Performance Management, Change Control, Network Protocols, Authentications, Network Monitoring, Multi-Factor Authentication, Key Management, Automation, Application Deployment, Cloud Services, Requirements Analysis, Business Continuity, Version Control, Continuous Deployment, Configuration Management, Continuous Integration

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FAQs

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What is the difficulty level of a cloud architect certification?

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High-quality free Cloud Architecture courses you can start today.

1. 1
   Computer Architecture
   Princeton UniversityAdvanced3 - 6 MonthsCourse4.7(3,872)
2. 2
   Amazon EC2 Basics
   Amazon Web ServicesBeginner1 - 4 WeeksCourse4.6(113)
3. 3
   Amazon Transcribe Getting Started
   Amazon Web ServicesBeginner1 - 4 WeeksCourse4.9(23)

Best Cloud Architecture professional programs

Earn a certificate in Cloud Architecture from top universities and companies.

1. 1
   AWS Cloud Solutions Architect
   Amazon Web ServicesIntermediate3 - 6 MonthsProfessional Certificate4.8(7,097)
2. 2
   Preparing for Google Cloud Certification: Cloud Architect
   Google CloudIntermediate3 - 6 MonthsProfessional Certificate4.7(61,787)
3. 3
   IBM Hybrid Cloud Architect
   IBMIntermediate3 - 6 MonthsProfessional Certificate4.6(7,823)

Article sources

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- Cybersecurity
- Data Analytics

- Digital Marketing
- Human Resources (HR)
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- Project Management
- Python
- SQL

Professional Certificates

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- Google Cybersecurity Certificate
- Google Data Analytics Certificate
- Google IT Support Certificate
- Google Project Management Certificate
- Google UX Design Certificate
- IBM AI Engineering Certificate
- IBM AI Product Manager Certificate
- IBM Data Science Certificate
- Intuit Academy Bookkeeping Certificate

Courses & Specializations

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- AI For Business Specialization
- AI For Everyone Course
- AI in Healthcare Specialization
- Deep Learning Specialization
- Excel Skills for Business Specialization
- Financial Markets Course
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